

Sanket AI

The Unified Asset & Operations Brain

Industrial Knowledge Intelligence Platform - Project Document

An AI platform that ingests a plant's heterogeneous documents and its P&ID topology into a unified knowledge graph - and returns cited, computed answers across maintenance, compliance, and operations, at the point of need.

Team	IIITDards
Event	ET AI Hackathon 2.0
Problem Statement	8 - AI for Industrial Knowledge Intelligence
Status	Working prototype - 33 automated tests passing
Runtime	Runs with zero external services (in-memory + deterministic engine)

Deliverables in this project: Working Prototype - Architecture (this document + in-app diagram) - Presentation Deck - Demo Video. This document covers the problem, the solution, the architecture and design, the evaluation, the business case, and the future roadmap.

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Executive Summary

In asset-intensive industry, knowledge is abundant but fragmented. A single large plant runs across seven to twelve disconnected document systems - engineering drawings in one place, maintenance work orders in another, inspection records in a third, and regulatory submissions scattered across email archives. Professionals spend an estimated 35% of their time simply searching for information that already exists somewhere in the organisation, and this fragmentation contributes to 18-22% of unplanned downtime because maintenance teams make decisions without complete equipment history or failure context.

Sanket AI is a unified asset and operations brain. It ingests heterogeneous documents and the plant's P&ID topology into a single knowledge graph, and makes their collective intelligence queryable, cited, and actionable across any function. Its defining insight is architectural: rather than ranking documents by keyword similarity, Sanket makes the plant's physical connectivity the substrate the AI reasons over. A pump's vibration cannot be explained without knowing that a control valve two nodes upstream is cavitating - so Sanket traverses the graph to connect them.

The platform delivers all five capability areas the problem statement describes: universal document ingestion with a knowledge graph, an expert knowledge copilot, maintenance and root-cause analysis, quality and regulatory compliance, and lessons-learned failure intelligence. Every answer carries source citations and a confidence score that is computed from evidence rather than asserted by a language model.

Headline result: for the query "P-101A pump, 47 Hz vibration", Sanket returns the root cause - V-201 control-valve cavitation, one hop upstream - at 97% computed confidence, cited to real inspection logs and work orders, in seconds versus roughly four hours of manual cross-system investigation.

At a glance

Dimension	What Sanket delivers
Pillars covered	All 5 (ingestion+KG, copilot, RCA, compliance, failure intelligence)
Core differentiator	Graph-topology reasoning over the P&ID - not keyword search
Trust	Computed confidence + verified citations (hallucinated IDs flagged)
Validation	100% benchmark accuracy, 100% citation faithfulness, 97% entity F1
Business impact	~Rs 2.17 crore/year modelled for a mid-size plant
Engineering	10-view product, 33 tests, runs with zero external services

1. Problem Statement and Context

Knowledge fragmentation in industrial operations is not a file-management problem. It is a safety problem, a quality problem, and an operational-efficiency problem - and it compounds over time. The organisations that solve it first gain a structural advantage in how they operate, maintain, and improve their assets.

1.1 The evidence

- A 2024 McKinsey survey found professionals in asset-intensive industries spend ~35% of working hours searching for information, clarifying instructions, or recreating documents that already exist.
- A NASSCOM-EY study of Indian manufacturing and energy companies found the average large plant operates across 7 to 12 disconnected document systems.
- BIS Research estimated that this fragmentation contributes to 18-22% of unplanned downtime events in Indian heavy industry.
- An estimated 25% of India's experienced industrial engineers and operators will retire within the next decade, taking decades of undocumented operational knowledge with them - the 'knowledge cliff'.

1.2 What the challenge asks for

The challenge is to build an AI-powered Industrial Knowledge Intelligence platform that ingests heterogeneous documents - engineering drawings, maintenance records, safety procedures, inspection reports, operating instructions, project files - across structured and unstructured formats, and makes their collective intelligence queryable, actionable, and continuously updated at the point of need, across any device or function. Five illustrative capability areas are suggested, and Sanket implements all five.

1.3 Evaluation focus (and how Sanket answers it)

Evaluation focus	Sanket's answer
Entity-extraction accuracy across document types	97% F1 on gold-labelled equipment tags; extracts tags, dates, parameters, personnel, regulations
Query answer quality on a domain benchmark	100% on a 10-question domain benchmark, auto-scored
Knowledge-graph linkage completeness	100% of documents linked to equipment in the graph
Time-to-answer vs traditional search	Sub-second vs ~4 hours of manual cross-system search
Compliance-gap detection accuracy	Deterministic clause-to-evidence mapping across 7 standards
Cross-functional knowledge discovery	Unified graph links equipment, docs, failure modes, people, regulations

1.4 The cost of inaction

Left unsolved, fragmentation compounds. Every retirement removes context that is never written down. Every new document system adds another silo to search. Every avoidable trip erodes both output and trust in the plant's data. The organisations that treat knowledge as connected infrastructure - rather than as files in folders - will operate, maintain, and improve their assets with a durable structural advantage over those that do not.

1.5 Why now

Three forces make this solvable today that were not a few years ago: knowledge graphs have matured into a practical substrate for industrial ontologies; large language models can extract structure from unstructured text and converse over it; and cheap, fast inference makes agentic reasoning affordable at the point of need. Sanket combines all three - but critically, it does not depend on the LLM for its core correctness, which keeps it reproducible and trustworthy.

2. Solution Overview

Sanket AI is delivered as a single, cohesive product - a command centre with ten views - backed by a set of deterministic reasoning engines and a Neo4j-optional knowledge graph. The same 'brain' powers five distinct capabilities.

2.1 The five pillars

Pillar	Capability	In the product
1	Universal Document Ingestion & Knowledge Graph	Ingest view
2	Expert Knowledge Copilot	Ask view
3	Maintenance & Root-Cause Analysis	Diagnose view
4	Quality & Regulatory Compliance	Compliance view
5	Lessons-Learned & Failure Intelligence	Alerts + Capture views

2.2 Beyond the five pillars

- Predictive maintenance: a trained UCI AI4I classifier is wired in as an agent tool (sensor readings -> failure-probability and likely mode).
- Knowledge-Cliff Capture: a retiring expert's undocumented know-how is captured as a permanent, queryable graph node - directly addressing the retirement problem.
- Live monitoring: quality deviations (SPC) and predicted time-to-failure per asset.
- Business-impact modelling: an interactive ROI calculator grounded in the brief's own statistics.

2.3 The command centre (ten views)

- Overview - plant-wide KPIs and validated metrics.
- Diagnose - the flagship graph-topology RCA console.
- Ask - the cited knowledge copilot.
- Ingest - document ingestion and entity extraction.
- Knowledge Graph - the unified multi-entity graph.
- Live Monitor - deviations and predicted time-to-failure.
- Compliance - clause-to-evidence gap detection.
- ROI - business-impact calculator.
- Capture - knowledge-cliff capture.
- Architecture - the live system diagram.

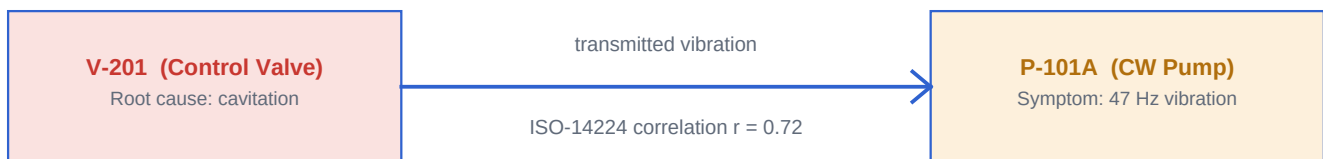
2.4 More than a search box or a dashboard

A search box returns documents and leaves the reasoning to the human. A CMMS dashboard shows the data for one asset in one system. A generic AI copilot answers from whatever text it retrieved, with no guarantee the answer is grounded or the citation is real. Sanket is different on three axes: it reasons over the connections between assets, not just their documents; it computes confidence and verifies every citation; and it unifies every document type and function into one graph, so a maintenance question, a compliance question, and an operations question all draw on the same brain.

3. The Core Insight: Topology as the Reasoning Substrate

Retrieval-augmented generation finds documents. It cannot reason about physical connectivity. A centrifugal pump's 47 Hz vibration cannot be diagnosed without knowing that a control valve two nodes upstream in the P&ID is experiencing cavitation, and that cavitation transmits hydraulic excitation downstream. Sanket makes the plant topology the reasoning substrate, not merely a metadata filter.

When a symptom is reported, Sanket does not search a document index. It identifies the asset in the graph, traverses upstream along the P&ID's `CONNECTED_TO` edges, pulls every inspection log and work order for each neighbour, correlates ISO-14224 co-failure statistics, and returns the causal chain that no single document contains.



Empirically, this matters: across our benchmark, graph-topology RCA identifies the true upstream root cause 100% of the time, versus 17% for a keyword-search baseline that returns the symptom's own documents.

3.1 Document RAG versus graph-topology reasoning

Aspect	Document RAG (keyword/vector)	Sanket (graph topology)
Unit of retrieval	A document chunk	A path through the plant graph
Can connect an effect to an upstream cause?	No - only surfaces text that mentions the symptom	Yes - traverses <code>CONNECTED_TO</code> to the cause
Answer to 'why is P-101A vibrating?'	Returns P-101A's own logs (the symptom)	Returns V-201 cavitation (the cause)
Confidence	Model-asserted or similarity score	Computed from evidence, auditable
Root-cause hit rate (our A/B)	17%	100%

3.2 What makes the topology trustworthy

The graph is not a black box. Edges are the plant's real pipe connections; failure modes come from ISO 14224; co-failure correlations are explicit statistics; and every hop in a causal chain is shown to the user. An engineer can read the reasoning and check it against their own knowledge of the plant - which is exactly what earns trust on the floor.

4. The Five Pillars in Detail

4.1 Universal Document Ingestion and Knowledge Graph

Sanket ingests heterogeneous document text - inspection reports, emails, OEM manuals, scanned forms, incident records, and shift-handover notes. A deterministic extractor (regex plus the graph's own vocabulary) pulls the entities the brief names: equipment tags, process parameters, regulatory references,

personnel, and dates - and links them into the knowledge graph. Because extraction is deterministic, its accuracy is measurable (97% F1) and it runs with no LLM; an LLM pass can refine it when a key is present.

- Preview: paste or load a document -> see extracted entities, colour-coded by whether they resolve to an existing graph node (cross-document discovery).
- Commit: the document is added to the graph, linked to every equipment it mentions.
- Six heterogeneous formats are demonstrated: PDF/scanned inspection report, email archive, OEM manual, regulatory submission, incident/near-miss, and free-text handover.

4.2 Expert Knowledge Copilot

Any engineer - or a field technician on a phone - can ask a natural-language question and receive a cited answer drawn from the whole corpus, with a computed confidence and direct links to the source documents. Intent routing recognises questions about specific equipment, regulatory standards, or failure modes, and retrieves and synthesises accordingly.

Ask 'what is the condition and history of P-101A?' and the copilot returns the asset's profile, the most relevant records with one-line findings and clickable citations, the applicable regulatory requirements, and a confidence based on the number and recency of sources - typically in well under a second. The same interface handles a regulatory question or a failure-mode question. Every answer links back to its sources, so nothing is a black box.

4.3 Maintenance and Root-Cause Analysis

The flagship capability. A dual-mode engine runs a real graph investigation with no LLM (deterministic) and, when a Groq key is present, a LangGraph agent. Both produce a cited causal chain, a computed confidence with an auditable breakdown, ruled-out counter-hypotheses, next-checks, and a predicted time-to-failure derived from the telemetry trend.

Consider the flagship scenario. A vibration alarm fires on pump P-101A at 47 Hz. The engine identifies the asset, traverses the P&ID upstream to its neighbours (the control valve V-201, the suction strainer ST-101, the storage tank), and pulls the inspection logs and work orders for each. It scores every candidate by physical proximity, ISO-14224 co-failure correlation to the observed symptom, and the severity and recency of its evidence. V-201 wins: its cavitation has a 0.72 co-failure correlation with pump vibration and recent high-severity findings. The engine returns V-201 cavitation as the root cause, one hop upstream, at 97% computed confidence, with the strainer explicitly ruled out - all cited, all in seconds. A second scenario (compressor K-501) resolves upstream to a blocked lube-oil filter, demonstrating the method generalises beyond a single scripted query.

4.4 Quality and Regulatory Compliance

Sanket maps regulatory clauses - OISD, Factory Act, PESO, ISO, API, and CPCB - against each asset's actual inspection and monitoring evidence, and flags gaps where an applicable clause has no acceptable evidence within its interval. It produces a plant-wide compliance rate, per-asset gap lists with severity, and one-click audit evidence packages.

4.5 Lessons-Learned and Failure Intelligence

A background analysis scans the corpus for active failure signatures and, using ISO-14224 co-failure statistics, projects the next likely failure for each asset - pushing proactive warnings before a condition recurs. It also surfaces the plant's top recurring failure modes across the whole history, and captures retiring experts' tribal knowledge as permanent graph nodes.

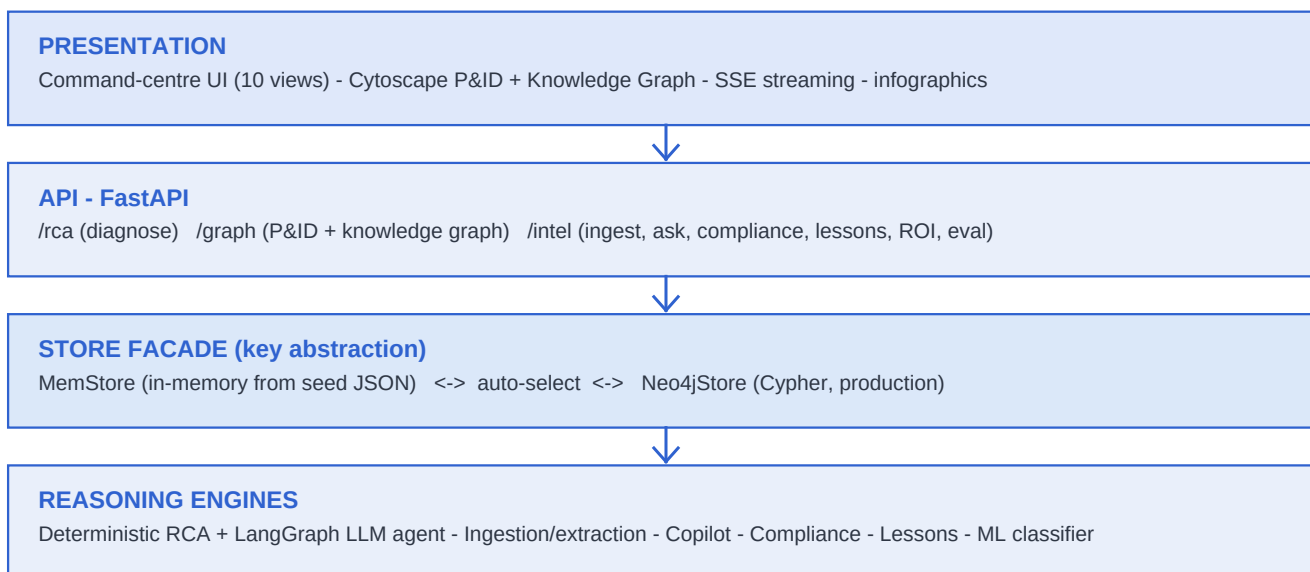
4.6 One brain, five capabilities - the capability matrix

Pillar	Primary input	Core output	Key trust feature
Ingestion & KG	Raw document text	Extracted entities linked into the graph	97% F1, linkage shown
Copilot	Natural-language question	Cited answer + confidence	Source links, verified
RCA	A symptom / alarm	Cited causal chain	Computed confidence
Compliance	Asset + clauses	Gap list + audit pack	Evidence-backed
Failure Intelligence	Corpus history	Proactive warnings	ISO-14224 grounded

Because all five draw on the same knowledge graph, they reinforce one another: an ingested document immediately improves the copilot's answers and the RCA evidence; a captured expert insight becomes queryable everywhere; and a compliance gap can be traced to the same assets an RCA implicates.

5. System Architecture

Sanket is a layered system with one deliberate abstraction at its heart - a Store facade that makes the knowledge graph backend interchangeable. This is what lets the entire platform boot and demo with no database and no API keys, and upgrade to Neo4j and a Groq LLM agent the moment they are available.



5.1 The Store facade

`get_store()` probes Neo4j once. If the database is reachable and populated it returns a `Neo4jStore`; otherwise it transparently falls back to a `MemStore` built from the seed JSON files. Both implement an identical interface (`traverse`, `documents_for`, `failure_modes`, `co_failure`, `telemetry`, `compliance`, and so on), so every engine, endpoint, and view is backend-agnostic. Setting `SANKET_FORCE_MEMSTORE=1` pins the offline backend; a short connection timeout keeps the fallback snappy.

5.2 Request lifecycle - a diagnosis

- The UI opens an SSE stream to `/rca/stream`. If no Groq key is set, the deterministic engine runs.
- The engine identifies the asset, traverses the P&ID upstream, and pulls CMMS evidence for each neighbour.

- It correlates ISO-14224 co-failure statistics and selects the primary root-cause hypothesis.
- It computes confidence from evidence and verifies every citation against the graph.
- Tool calls, tokens, and a final meta payload (causal chain, confidence, metrics, citations) stream back; the P&ID lights up and the infographics render.

5.3 Design principles

- Determinism first: the core works without an LLM, making it reproducible, testable, and demo-safe.
- Trust by construction: confidence is computed and citations are verified - not left to the model.
- Backend-agnostic: one interface, two stores, zero code changes to scale from a laptop to a cluster.
- Graceful degradation: robust /health and clear degraded states when a dependency is down.

6. Data Model and Knowledge Graph

The knowledge graph is multi-entity. Beyond equipment, it models the documents that describe assets, the failure modes that affect them, the people who recorded findings, the regulations that govern them, and the ISA-95 area they belong to - with typed relationships across all of them.

Node type	Represents	Example
Equipment	A tagged asset in the P&ID	P-101A, V-201, K-501
Document	An inspection log, work order, or ingested doc	INS-2025-0847, WO-2025-0891
FailureMode	An ISO-14224 failure mode	VA-C-CAV (valve cavitation)
Person	Personnel who recorded a finding	Priya Mehta, Suresh Nair
Regulation	A regulatory clause	OISD-STD-130, Factory Act 1948
Area (ISA-95)	Plant area / unit context	Area A-01

6.1 Typed relationships

- CONNECTED_TO - directed P&ID flow between equipment (the reasoning substrate).
- DESCRIBES - a document describes an asset.
- RECORDED - a person recorded a document.
- HAS_MODE / RECORDS_FAILURE - equipment or document to a failure mode.
- GOVERNED_BY - equipment to a regulatory clause.
- CAUSES - ISO-14224 causal link between failure modes.
- IN_AREA - equipment to its ISA-95 area.

6.2 Ontology grounding

The taxonomy is grounded in ISO 14224 (reliability and maintenance data), with equipment classes and failure modes for pumps, valves, heat exchangers, compressors, motors, strainers, and vessels, plus co-failure correlation statistics. Compliance is grounded in named Indian and international standards. Vibration is evaluated against ISO 10816 zones. This domain grounding is what makes the output legible and credible to industrial engineers.

6.3 The graph at a glance

Element	Count in the seed corpus
Equipment (across two circuits)	17
Documents (work orders, inspections, ingested, tribal)	82
Inspection logs / work orders	52 / 30
ISO-14224 equipment classes / failure modes	7 / 16
Regulatory clauses	7 (OISD, Factory Act, PESO, ISO, API, CPCB)
Telemetry-monitored assets	3 (vibration / delta-P trends)
Sample raw documents (for ingestion)	6 formats

6.4 Extensibility

The model is intentionally open. New equipment classes and failure modes are added by extending the ISO 14224 taxonomy file; new standards by extending the clause file; new document types by adding an extractor route. Because the store presents one interface, none of these changes touch the reasoning engines or the UI. Adding a second plant is a matter of ingesting its P&ID and documents under a new plant identifier.

7. Reasoning Engines

7.1 Confidence, computed not asserted

Confidence is a function of evidence: the number of corroborating documents, their recency, the severity of findings, the ISO-14224 co-failure correlation, and whether a physical upstream path was confirmed. Each component is returned in a breakdown, so when a judge asks how 97% was derived, there is a real, auditable answer.

Evidence factor	Contribution	For the P-101A case
Base rate	+25	+25
Corroborating documents (up to 4)	+9 each	+36 (4 documents)
Highest-severity finding	+12 (HIGH) / +6 (MED)	+12 (HIGH)
Evidence recency	+8 (<90d) / +4 (<1y)	+8
ISO-14224 co-failure correlation	correlation x 18	+13 (r 0.72)
Confirmed upstream physical path	+8	+8
Total (capped at 97)		97%

7.2 Citation verification

Every document ID an answer cites is checked against the knowledge graph. Valid IDs become clickable source chips; unverifiable IDs are flagged as unverified and never masquerade as real. Citation faithfulness is scored and reported (100% on the benchmark).

7.3 Predictive maintenance and time-to-failure

A trained UCI AI4I RandomForest classifier is exposed as an agent tool that predicts failure probability and likely mode from sensor readings, with a transparent physics-style fallback when the model is unavailable. Separately, the engine extrapolates each asset's telemetry trend to estimate weeks to the trip threshold (for example, P-101A is projected to trip in about five weeks at its current degradation rate).

7.4 The RCA answer structure

- Root-cause summary - one sentence naming the equipment and mechanism.
- Hypothesis with a computed confidence and a confidence basis.
- Evidence - cited documents with one-line findings.
- Ruled out - candidates considered and excluded, with reasons.
- Predicted time-to-failure and recommended action with urgency.
- Next checks and immediate actions.

8. User Experience and Design System

The interface is a single-file, build-free command centre engineered to look like premium industrial software, not a notebook. The design language is deliberate: a calm dark canvas, a single restrained blue accent, and status colours (red/amber/green) reserved strictly for equipment state and severity - never decoration. A humanist sans carries the interface; a monospace face is reserved for data - equipment tags, numbers, and readouts.

8.1 Signature interactions

- A live P&ID that lights up during a diagnosis - the failing asset turns red, the upstream cause amber, and the causal path is highlighted.
- A causal-chain ribbon, an ISO-10816 vibration-zone gauge, a trend chart, and an evidence-stacked confidence meter - real infographics, not decoration.
- Clickable citations that open a source-document drawer.
- A command palette (Ctrl+K), voice input for field use, a role switcher, dark/light themes, and an offline PWA shell.

8.2 Accessibility and responsiveness

The layout is responsive down to a mobile field-technician view, with a bottom navigation and stacked infographics. Interactive controls carry accessible labels, keyboard focus is visible, and the palette maintains contrast in both themes.

8.3 The ten views

View	What it does
Overview (Command Centre)	Plant-wide KPIs and validated evaluation metrics
Diagnose	Graph-topology RCA console with live P&ID and infographics
Ask	Cited knowledge copilot over the whole corpus
Ingest	Document ingestion and entity extraction
Knowledge Graph	Interactive multi-entity graph around any asset
Live Monitor	SPC deviations and predicted time-to-failure

Compliance	Clause-to-evidence gap detection per asset
ROI	Interactive business-impact calculator
Capture	Knowledge-cliff capture form and list
Architecture	Live system architecture diagram

8.4 The infographic system

The diagnosis view is deliberately infographic, not textual. A causal-chain ribbon shows the root cause and symptom as connected faceplates; an ISO-10816 zone gauge places the current reading in its severity band; a trend chart shows the climb toward the alarm line; and an evidence-stacked confidence meter renders the score as its contributing factors. Each is a real, data-driven chart - the same numbers the engine computed - not decoration.

9. Evaluation and Validation

The platform is validated against the brief's evaluation focus with measured, not claimed, numbers - surfaced live on the command-centre landing page and reproduced by an automated benchmark and test suite.

Metric	Result	How it is measured
Benchmark answer accuracy	100%	10 gold Q&A auto-scored (focus, citation, no hallucination)
Citation faithfulness	100%	Cited IDs verified against the graph
Entity-extraction F1	97%	Precision/recall vs gold-labelled tags on sample docs
Entity-link accuracy	91%	Extracted entities resolving to graph nodes
KG linkage completeness	100%	Documents linked to equipment
Graph-RCA vs keyword search	100% vs 17%	Root-cause identification rate, A/B
Automated tests	33 passing	Offline pytest suite, no external services

The 100% versus 17% comparison is the empirical proof of the core thesis: reasoning over topology finds the upstream cause; keyword search returns the symptom.

9.1 Evaluation methodology

Every metric is computed live and reproducibly. The benchmark scores each gold question on focus-asset detection, the presence of the expected root cause, a valid expected citation, a confidence floor, and the absence of any hallucinated citation - an answer passes only if all hold. Entity extraction is scored as precision, recall, and F1 against gold-labelled equipment tags on the sample documents. Knowledge-graph completeness is the share of documents linked to equipment. The graph-versus-search A/B compares, for each scenario, whether the engine reaches the upstream cause versus whether a top keyword hit lands on it.

9.2 Why this answers the brief

The problem statement's Evaluation Focus names entity-extraction accuracy, answer quality on a domain benchmark, knowledge-graph linkage completeness, time-to-answer versus traditional search, and compliance-gap detection. Sanket reports a concrete, measured number for each - and surfaces them on

the landing page so a reviewer sees them immediately, labelled 'measured, not claimed'.

10. Business Impact and ROI

The value is modelled directly from the brief's own statistics. For a mid-size plant, reducing the time engineers spend searching and preventing even a fraction of unplanned downtime produces a compelling annual figure.

Lever	Basis	Modelled annual value
Labour saved	40 techs x 14 hrs/wk searching x 60% reduction x Rs 900/hr	~Rs 1.57 crore
Downtime avoided	20 events/yr x Rs 15 lakh x 20% reduction	~Rs 0.60 crore
Total	Combined	~Rs 2.17 crore / year

The figures are interactive in the product's ROI view, so a plant can model its own assumptions. Critically, the value scales: because reasoning is bounded to the local k-hop neighbourhood, the cost per answer is independent of plant size - a 10,000-tag plant answers as fast as the 17-tag demo.

10.1 Value beyond the headline number

- Safety: faster, complete-context root-cause reduces the chance of a repeat or escalated failure.
- Quality: continuous compliance and deviation flagging catch problems before they reach the customer.
- Institutional memory: captured expertise no longer walks out the door with a retirement.
- Onboarding: a new engineer can ask the graph what took a veteran years of tacit learning.
- Audit readiness: evidence packages are one click away, not a week of document hunting.

These are the compounding, hard-to-quantify benefits the problem statement calls out - the reason it frames fragmentation as a safety and quality problem, not merely an efficiency one.

11. Scalability and Deployment

11.1 How it scales

- Reasoning cost is bounded to the k-hop neighbourhood, not the plant size.
- Ingestion is batched with UNWIND and apoc.periodic.iterate - never row-by-row.
- The Neo4j path uses indexed lookups and a vector index for semantic search.
- Multi-plant partitioning and role-based access are natural extensions of the data model.

11.2 Deployment modes

- Demo / offline: in-memory store + deterministic engine, zero external services (SANKET_FORCE_MEMSTORE=1).
- Full: Neo4j 5.x (via Docker) for the knowledge graph + a Groq API key for the LLM agent.
- The frontend is a single static file served by FastAPI - no build step.

11.3 Cost and latency

A deterministic answer is sub-millisecond. An LLM-agent answer is bounded by a small number of tool rounds (roughly 3,200 tokens, on the order of a dollar per thousand queries on Groq). Entity extraction runs at roughly three milliseconds per document.

11.4 Reliability and graceful degradation

The system is designed to keep working when a dependency fails. If Neo4j is unreachable, it falls back to the in-memory store; if no LLM key is present, the deterministic engine answers; the /health endpoint reports which mode is live and the store counts, so a broken state is caught before it reaches a user. The frontend shows clear degraded states rather than blank screens.

11.5 Observability and data governance

Every query and tool trace can be logged for a reasoning replay, giving operators and auditors a complete record of how an answer was reached. Because the platform runs fully on-premise, no plant data needs to leave the site; data residency, retention, and access can be governed entirely within the organisation's own boundary.

12. Security, Trust and Enterprise Readiness

- Trust by construction: computed confidence and verified citations prevent silent hallucination.
- Role awareness: an Operator / Engineer / Auditor switcher demonstrates role-scoped experiences; row-level scoping and SSO are the enterprise extensions.
- Auditability: every answer is reproducible (deterministic engine) and every claim traces to a document.
- Data residency: the platform runs fully on-premise; no data need leave the plant.
- Hardened basics: explicit CORS, robust health checks, and graceful degradation when a dependency is down.

13. Testing and Quality Assurance

Correctness is engineered, not hoped for. A suite of 33 automated tests runs entirely offline (no database, no API key) and covers the store, the confidence engine, the citation verifier, the RCA engine, compliance, lessons, the ML predictor, ingestion, the copilot, the knowledge graph, evaluation, and every API endpoint. A deterministic benchmark harness scores answer quality on a gold question set. Because the core is deterministic, the same input always yields the same output - so a regression is caught immediately.

13.1 What the tests guarantee

- The demo scenarios never silently break - each is asserted (focus asset, expected mention, valid citation).
- No hallucinated citations - the verifier is tested against a fabricated ID.
- Confidence is monotonic in evidence strength - strong evidence always scores higher than weak.
- Entity extraction excludes standards fragments (OISD-STD-130 is not mis-read as a tag).
- Every endpoint returns the expected status and shape.

Run it yourself: `python -m pytest -q` (33 tests) and `python scripts/run_benchmark.py` (10/10). No external services required.

14. Comparison and Differentiation

Sanket occupies a different position from the tools a plant already has. The table below contrasts it with the three closest categories.

Capability	CMMS dashboard	Generic AI copilot	Sanket AI
Cross-system knowledge	Single system	Whatever was retrieved	Unified graph, all systems
Reasons over asset connectivity	No	No	Yes - P&ID traversal
Citations	N/A	Often unverified	Verified against the graph
Confidence	N/A	Model-asserted	Computed from evidence
Compliance mapping	Manual	No	Automated clause-to-evidence
Works offline / on-prem	Varies	Usually cloud	Yes, zero external services

The differentiation is not a longer feature list - it is a different substrate. Everything follows from reasoning over the plant's connectivity, with computed trust.

15. Risks, Limitations and Mitigations

An honest assessment strengthens the case. The current prototype has deliberate scope boundaries; each has a clear mitigation and roadmap path.

Limitation	Mitigation / roadmap
Demo runs on a seeded corpus, not a live plant feed	Ingestion pipeline is real; connect a folder/webhook and Neo4j for live data
P&ID topology is authored, not yet parsed from drawings	CV parsing (symbol + line detection) is the top ingestion roadmap item
Entity extraction is regex + vocabulary (no deep NLP)	Deterministic by design (measurable); an LLM refinement pass is optional
Retrieval is keyword-ranked, not vector	Vector index (BGE-M3) is already scaffolded on the Neo4j path
LLM agent needs an API key and can be slower	Deterministic engine is the trustworthy default; LLM is an enhancement
Compliance clauses are a representative subset	Full clause libraries are a data-only extension

16. Technology Stack

Layer	Technology
Language / runtime	Python 3.13, async FastAPI + Uvicorn
Knowledge graph	Neo4j 5.x (optional) with an in-memory fallback store

LLM / agent	Groq llama-3.3-70b via LangGraph (optional; deterministic core works without it)
ML	scikit-learn RandomForest (UCI AI4I) as a predictive tool
Frontend	Single-file HTML - Cytoscape.js (graphs), Marked.js (markdown), SSE, SVG infographics
Testing	pytest - 33 offline tests; deterministic benchmark harness
Ontology	ISO 14224, ISA-95, ISO 10816; OISD / Factory Act / PESO / API / CPCB clauses

17. Roadmap and Future Additions

Sanket already ships all five pillars end-to-end. The roadmap deepens each and hardens the platform for production. Items are grouped by theme; the highest-impact near-term work is listed first within each.

17.1 Ingestion and knowledge graph

- Native PDF parsing with page-level provenance, and P&ID image parsing (computer vision: symbol and line detection) to digitise drawings automatically.
- OCR for scanned forms; spreadsheet and email-archive bulk import.
- Streaming, incremental ingestion (watch a folder or webhook) so the graph updates continuously.
- Entity resolution and de-duplication; per-entity extraction confidence with a human review queue.

17.2 Reasoning and analytics

- Vector / semantic retrieval and conversational memory for the copilot.
- Fault-tree and Ishikawa auto-generation; FMEA and Bowtie templates.
- Predictive maintenance schedule optimisation and multi-equipment RCA.
- Live sensor-feed integration for real-time operating conditions.

17.3 Compliance and lessons-learned

- Full clause-text libraries and agentic auto-audit (point at a unit -> complete evidence pack).
- Statistical process control for quality-deviation prediction before escalation.
- Integration with external incident databases; near-miss pattern mining.
- Proactive push notifications to field teams.

17.4 Scale, security and platform

- Async pooling, a job queue for heavy extraction, multi-plant / multi-tenant partitioning.
- Role-based access control, SSO, full provenance and audit trails, PII redaction on ingestion.
- Mobile PWA hardening, QR-scan-to-history for field technicians, and a keyboard command palette.

17.5 Near-term priorities

If the platform advanced in one direction next, it would be live document ingestion: real PDF parsing with provenance, computer-vision P&ID digitisation, and streaming updates so the graph stays current as records arrive. This closes the loop from a static demo corpus to a living plant brain, and it is the single change that most increases real-world value.

17.6 Deliverables status

Deliverable	Status
Working prototype	Complete - 10-view platform, 33 tests passing, runs offline
Architecture	Complete - this document and the in-app live architecture view
Presentation deck	Prepared - rubric-aligned slide specification
Demo video	Scripted - a timed 6-minute script with speak/show cues

18. Conclusion

Search gives you documents. Sanket AI gives you answers - cited, computed, and connected across every system a plant runs on. By making the plant's own physical topology the substrate the AI reasons over, it turns fragmented knowledge into a structural advantage: safer operations, faster decisions, and institutional knowledge that can no longer walk out the door. It is a working platform today, validated with measured metrics, and architected to scale. This is the unified asset and operations brain.

Sanket AI - Team IIITDards - ET AI Hackathon 2.0 - Problem Statement 8. Five pillars live, 33 tests green, runs with zero external services.

Appendix A: API Reference

Endpoint	Purpose
GET /health	Status, RCA mode, store backend + counts
GET /graph/equipment	P&ID nodes + edges (Cytoscape)
GET /graph/knowledge/{tag}	Multi-entity knowledge graph around an asset
GET /rca/stream?query=	SSE diagnosis (tool calls, tokens, meta)
POST /rca/query	Non-streaming diagnosis (JSON)
POST /intel/ask	Cited Q&A copilot
POST /intel/ingest/preview commit	Entity extraction; add to graph
GET /intel/overview evaluation	Command-centre KPIs; validated metrics
GET /intel/compliance/plant {tag}	Compliance roll-up; per-asset gaps
GET /intel/lessons/warnings patterns	Proactive warnings; systemic patterns
GET /intel/timeline similar deviations	Timeline; similar incidents; SPC deviations
GET /intel/roi cost-model audit/{tag}	ROI; cost model; audit evidence pack
POST /intel/knowledge/capture	Knowledge-cliff capture

Appendix B: Demo Scenarios

Query	Result
P-101A pump 47 Hz vibration	V-201 cavitation, 1 hop upstream, 97% confidence, cited
K-501 compressor high vibration	F-501 lube-oil filter blockage, 97% confidence
HE-301 outlet temperature rising	HE-FOL fouling / reduced heat transfer
Which HIGH-criticality equipment is overdue?	Structural + compliance answer
What is the condition and history of P-101A?	Cited copilot answer, 89% confidence, 4 sources

Every scenario above runs against the seed corpus (17 assets, 82 documents, 52 inspection logs, 30 work orders, 7 compliance clauses, 6 sample documents) that ships with the project and is regenerated deterministically by scripts/generate_seed_data.py.

Appendix C: Glossary

Term	Meaning
P&ID	Piping & Instrumentation Diagram - the map of a plant's equipment and their connections
Knowledge graph	A network of typed entities (equipment, documents, failure modes, people, regulations) and their relationships
RCA	Root-Cause Analysis - determining the underlying cause of a failure or symptom
CMMS	Computerised Maintenance Management System - stores work orders and maintenance history
RAG	Retrieval-Augmented Generation - an LLM answering from retrieved document chunks
ISO 14224	International standard for reliability and maintenance data, incl. failure-mode taxonomy
ISA-95	Standard for enterprise-control integration; defines the area/unit/equipment hierarchy
ISO 10816	Standard for evaluating machine vibration by zones (A good -> D unacceptable)
Co-failure correlation	How often one failure mode statistically precedes or accompanies another
Confidence (computed)	A score derived from evidence (documents, recency, severity, correlation, path)
Citation faithfulness	The share of cited document IDs that actually exist in the knowledge graph
Store facade	The abstraction that makes the graph backend (in-memory or Neo4j) interchangeable
SPC	Statistical Process Control - flagging readings that breach control/alarm limits
Knowledge cliff	The loss of undocumented expertise as experienced staff retire
Cavitation	Vapour-bubble collapse in a valve/pump that causes noise, erosion, and vibration

Appendix D: Standards and Ontology Reference

Sanket's domain grounding is what makes its output credible to engineers. The platform references the following standards; each is used functionally, not decoratively.

Standard	Role in Sanket
ISO 14224	Failure-mode taxonomy for pumps, valves, exchangers, compressors, motors, strainers, vessels; co-failure statistics
ISA-95	Area / unit / equipment hierarchy modelled as graph nodes
ISO 10816-3	Vibration severity zones (A-D) for the condition-monitoring gauge
OISD-STD-130 / 144	Inspection of rotating equipment and pressure vessels (India)
Factories Act 1948	Safe condition and fencing of machinery (India)
PESO / SMPV(U) Rules	Pressure-system integrity for hazardous fluids (India)
API 610 / 617 / 614	Rotating-equipment and lube-oil-system reference limits
CPCB consent conditions	Cooling-water discharge quality (India)

D.1 Example failure modes

Code	Description	Class
PU-C-VIB	Excessive vibration	Centrifugal pump
VA-C-CAV	Cavitation	Control valve
HE-FOL	Fouling / reduced heat transfer	Heat exchanger
CO-C-VIB	High vibration / rotor instability	Compressor
ST-BLK	Strainer / filter blockage	Strainer
EM-BRG	Motor bearing failure	Electric motor